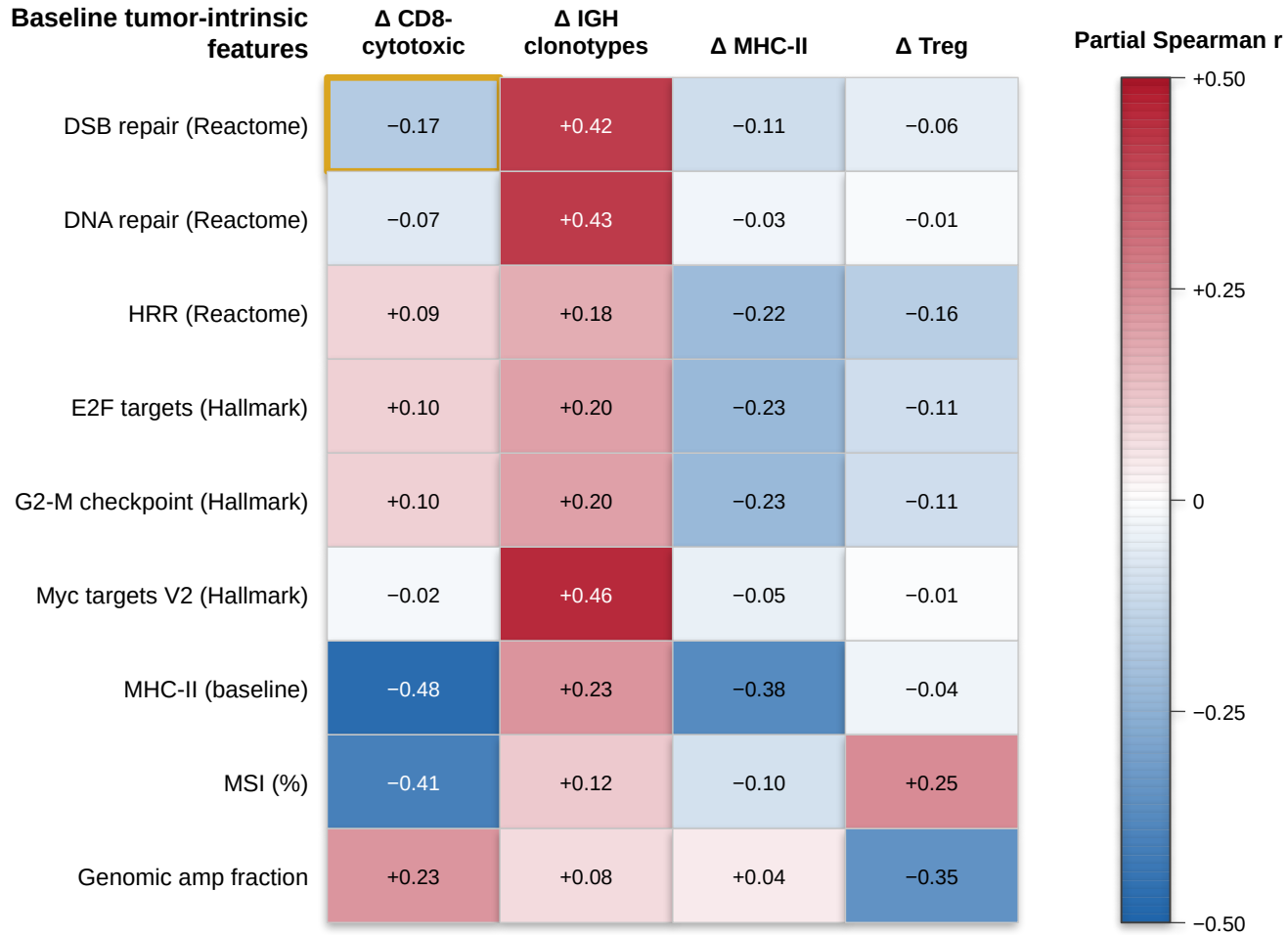


Cascade Δ features (radiation-phase dynamics)



n = 11–12 paired subjects · partial Spearman, response-group adjusted · BH-corrected across 36 pairs

Cell value = partial Spearman r. * nominal $P < 0.05$ ** BH $q < 0.05$ Gold border: headline pair (DSB repair $\times$ Δ CD8-cytotoxic).

Convergence result: 0/36 partial $P < 0.05$ · 0/36 BH $q < 0.05$ (static and dynamic layers observationally independent).